

AI-mediated authentic assessment and metacognitive reflection: A mixed-methods study of the AAIWA model

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ABSTRACT

This study introduces the Authentic AI-Assisted Writing Assessment (AAIWA) model, integrating authentic rubric-based assessment, condition-responsive AI feedback, and structured metacognitive reflection as a unified recursive architecture within Indonesian pre-service teacher education. Existing models treat these components additively, creating a critical design gap particularly in underrepresented Global South contexts. AAIWA addresses this by computationally embedding rubric criteria within the feedback engine and positioning structured reflection as a dynamic input into subsequent feedback cycles rather than a post-hoc add-on. The AAIWA prototype is a constrained hybrid system: a rule-based NLP layer governs all pedagogically consequential decisions while GPT-3.5-turbo serves exclusively as a language-articulation module. Claims of 'condition-responsive feedback' therefore denote rule-conditioned responsiveness, not autonomous LLM reasoning. A mixed-methods R&D design engaged 75 pre-service teachers across three universities in four iterative writing cycles. Significant improvements were observed in academic writing performance ($d = 0.92$, 95% CI [0.65, 1.19]) and metacognitive reflection ($d = 0.98$, 95% CI [0.69, 1.27]). A moderate writing reflection correlation ($r = .54$, 95% CI [.36, .68]) offers partial mechanistic support for the integrated design. Thematic analysis ($n = 15$) identified three themes: enhanced metacognitive awareness, motivational benefits of immediate feedback, and progressive internalisation of reflection. Expert validation yielded strong content validity indices (rubric: CVI = 0.84; inventory: CVI = 0.82). These pilot findings associated with rather than causally attributable to AAIWA indicate a theoretically grounded, empirically promising framework warranting controlled replication across diverse educational contexts.

1. Introduction

The integration of Artificial Intelligence in higher education has created both opportunities and challenges for writing assessment, particularly within teacher education. Traditional modes of assessment are widely criticized for failing to capture the complexity of academic writing, which requires not only linguistic competence but also higher-order cognitive and metacognitive skills. Pre-service teachers must develop capacity as reflective practitioners who can guide future generations; the stakes for designing effective, authentic, and technologically supported writing assessments are therefore especially high [1,2].

Authentic assessment offers a promising foundation. Defined as aligning tasks with real-world practices, it emphasizes performance, knowledge integration, and reflective judgment rather than rote recall [3]. Portfolios, project-based assignments, and reflective journals are

congruent with the complexity of professional communication [4]. Yet authentic assessment is resource-intensive, and its scalability remains a pressing concern creating a need for innovative solutions.

AI-driven writing feedback systems offer one such solution. Advances in NLP and large language models (LLMs) have enabled tools that provide real-time, adaptive feedback. AI-generated feedback can match or surpass teacher feedback in clarity and timeliness [1], and significantly improve argumentative writing in EFL contexts [5]. Concerns persist, however: AI may promote formulaic writing [6] and raise academic integrity risks [2]. Effective use therefore requires careful design and critical human oversight. Alongside the cognitive dimensions of writing, metacognitive processes are equally central. Embedding structured reflection within feedback loops enhances self-regulated learning [7] and writing self-efficacy among pre-service teachers [8].

The Indonesian context is theoretically significant beyond

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geographic representation. Three structural conditions constitute a distinct implementation environment: (a) infrastructural disparities between Java-based and outer-island institutions constraining cloud-based AI deployment; (b) a teacher-centered pedagogical tradition in which student self-assessment is often undervalued; and (c) an EFL context in which pre-service teachers compose academic texts in a second or third language, requiring additional cognitive processing for planning, monitoring, and linguistic formulation [9,10].

The present study designs, validates, and pilots the AAIWA model a mixed-methods R&D pilot grounded in a genuinely recursive architecture that distinguishes it from conventional additive frameworks. Unlike existing AWE systems, AAIWA computationally embeds rubric dimensions within the feedback engine and positions structured reflection as a dynamic input recalibrating subsequent cycle. The study makes three contributions: (1) the first architecturally integrated model that computationally unifies authentic rubrics, condition-responsive AI feedback, and metacognitive reflection within a recursive system; (2) preliminary empirical evidence of feasibility in a Global South teacher education context; and (3) a validated instrumentation suite the Authentic Writing Rubric (AWR) and Metacognitive Reflection Inventory (MRI) available for adoption in future controlled replications.

Despite substantial advances in authentic assessment, AI-assisted writing evaluation, and metacognitive reflection research, these three strands have largely developed in parallel rather than as an integrated assessment architecture. Authentic assessment provides pedagogically meaningful evaluation but remains difficult to scale; AI-generated feedback offers scalability yet is often detached from authentic rubric criteria; and structured reflection supports self-regulated learning but is typically positioned as a post-hoc activity rather than a dynamic component of feedback generation. Consequently, a critical theoretical and design gap remains: the absence of a recursive framework that computationally integrates authentic rubric-based assessment, condition-responsive AI feedback, and metacognitive reflection as mutually constitutive processes. This gap is particularly salient in teacher education and underrepresented Global South contexts, where scalable yet pedagogically robust assessment systems are urgently needed. The AAIWA model was developed to address this gap by positioning authentic rubric criteria, AI-generated feedback, and metacognitive reflection not as sequential components but as computationally interdependent processes operating within a unified recursive system.

The central novelty of this study therefore lies not merely in combining these three components, but in the architectural mechanism by which they are integrated: rubric criteria are computationally embedded within the feedback engine, and learner reflection dynamically recalibrates subsequent feedback priorities a design logic absents from existing AWE frameworks.

Three research questions guide the study:

RQ1 (Qualitative & Technical): How does the AAIWA model operationalize the integration of authentic rubrics, condition-responsive AI feedback, and structured metacognitive reflection as a unified assessment architecture?

RQ2 (Quantitative): What patterns of change in academic writing performance and metacognitive reflection scores are observed among pre-service teachers following AAIWA participation?

RQ3 (Qualitative): How do pre-service teachers perceive the role of AI-generated feedback and structured reflection in their writing development?

2. Background

2.1. Authentic assessment in higher education

Authentic assessment aligns tasks with real-world practices, requiring students to perform meaningful activities rather than recall isolated knowledge [3]. Portfolios, project-based learning, reflective

journals, and case studies promote transfer of knowledge beyond the classroom, enhancing agency and ownership of learning [4,11]. In teacher education specifically, authentic assessment cultivates reflective practice as a professional disposition essential for future teachers. The COVID-19 pandemic underscored the importance of adaptable, digitally mediated approaches; yet challenges persist: providing individualized, meaningful feedback at scale remains among higher education's greatest difficulties [12]. This tension creates the opportunity for AI integration examined in the following section.

2.2. AI-driven feedback and writing evaluation

AI-generated feedback significantly improved argumentative writing in EFL learners, surpassing teacher feedback in timeliness and consistency [1,5,13]. Against these benefits, Ravi and Besharat [6] warn of formulaic writing; García-López and Trujillo-Liñán [2] highlight academic integrity risks; and Banihashem et al. [14] found that AI feedback sometimes displaced deeper learning opportunities. Bearman et al. [15] conclude that AI systems must be transparent, fair, and embedded within reflective learning processes—conditions the AAIWA design directly addresses.

2.3. Metacognitive reflection and self-regulated learning

Metacognition thinking about one's own thinking is central to effective writing [16]. Sasson et al. [8] demonstrated that pre-service teachers with higher metacognitive awareness showed stronger writing self-efficacy and revision behaviour; Zhan [17] found that reflection prompts embedded in feedback cycles increased student engagement; and Zarestky et al. [7] provided large-scale evidence that reflective interventions enhanced metacognitive monitoring, leading to improved writing outcomes. In teacher education, reflection is not merely academic but a professional necessity: it forms the foundation of lifelong learning and effective pedagogy [11].

2.4. Integrating authentic assessment, AI feedback, and reflection

While previous research has documented the value of authentic assessment, AWE tools, and metacognitive reflection individually [18], integration often remains additive rather than systemic: rubrics are applied independently by instructors, AI feedback targets surface-level errors, and reflection is positioned as post-activity appraisal rather than a dynamic feedback input [19]. The AAIWA model addresses this by operationalizing three interrelated mechanisms: (1) rubric-embedded feedback generation authentic AWR dimensions are computationally mapped onto the feedback engine rather than applied externally; (2) condition-responsive iterative feedback adapts to persistent patterns across drafts, combining directive and metacognitive support; and (3) reflection-modulated recalibration student reflection inputs shape subsequent feedback priorities, so that learner metacognition and feedback generation operate reciprocally. Together, these create a recursive adaptive cycle that contrasts with conventional AWE designs in which reflective tasks remain external to the feedback generation process (see Fig. 1).

2.5. Research gap and conceptual framework

The three literature strands reviewed above authentic assessment (Section 2.1), AI-driven feedback (Section 2.2), and metacognitive reflection (Section 2.3) have each advanced considerably yet remain largely siloed in both theory and practice [18,19]. As Section 2.4 demonstrates, existing integration attempts remain additive rather than architecturally recursive: rubrics are applied externally, AI feedback targets surface errors independently, and reflection is positioned as post-activity appraisal rather than a dynamic feedback input. Critically, no existing framework has computationally embedded all three

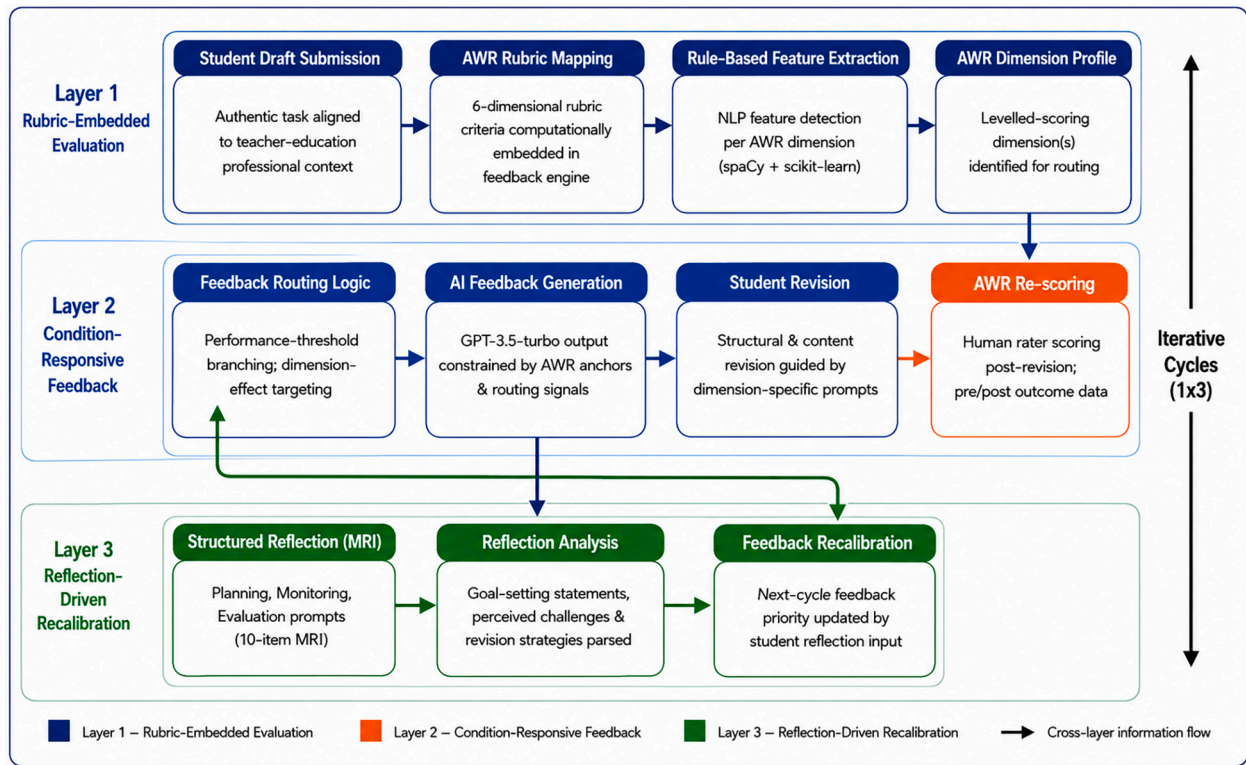


Fig. 1. The AAIWA condition-responsive assessment cycle.

Note. Shaded bands denote three system layers: Layer 1 – Rubric-Embedded Evaluation (dark blue); Layer 2 – Condition-Responsive Feedback (mid blue), with human re-scoring indicated by the orange hatched node; Layer 3 – Reflection-Driven Recalibration (green). ↓ = cross-layer information flow; ↻ = Layer 3 recalibration reprioritizes Layer 2 routing in the subsequent cycle; ↺ = recursive iteration across three writing cycles (1–3). AWR = Authentic Writing Rubric; MRI = Metacognitive Reflection Inventory; NLP = Natural Language Processing.

components within a single recursive system — particularly in Global South teacher education contexts where scalable yet pedagogically robust solutions are urgently needed [1,15,20]. Table 1 operationalizes these distinctions across three design configurations.

3. Methods

3.1. Research design

This study adopted a research and development (R&D) design combined with a mixed-methods approach to design, validate, and pilot AAIWA [22]. Three stages were followed: initial design of the prototype, expert validation of instruments, and small-scale pilot implementation. The mixed-methods approach captured both quantitative evidence (writing performance and reflection scores) and qualitative insights

(student perceptions), consistent with triangulation practices recommended in AI-in-education research [15,23].

3.2. Research context and participants

The study was conducted across three Indonesian universities offering teacher education programmes, purposively selected for their established academic writing curricula. A total of 75 pre-service teachers participated (25 per institution; 52 female, 23 male; aged 19–23), all enrolled in undergraduate teacher education programmes with foundational but developing academic writing skills. A priori power analysis (G*Power 3.1; $d = 0.60$, $\alpha = .05$, power = .80) established a minimum sample of $N = 23$; the enrolled $N = 75$ yielded estimated power $> .99$ for medium-to-large effects. All participants completed the full intervention (100% retention), with complete data available for all analyses.

Table 1

Operational distinctions between conventional AWE, AWE + reflection, and AAIWA.

Feature	Conventional AWE	AWE + Reflection	AAIWA
<i>Rubric Integration</i>	Not embedded; applied externally by instructor	Externally appended; post-hoc alignment	Computationally embedded within AI feedback engine; AWR dimensions operationalised as routing parameters
<i>Feedback Adaptability</i>	Minimal; generic rule-driven output	Limited; not dimension-sensitive	Adaptive; dimension-deficit targeting adjusts feedback type and depth across drafts
<i>Reflection Role</i>	Optional or absent	Post-task add-on; not integrated into feedback loop	Feedback-modulating input; MRI reflection recalibrates routing priorities for subsequent cycle
<i>Iterative Design</i>	Static; single-pass evaluation	Limited iteration; on-demand only	Recursive architecture across three writing cycles; each cycle informed by prior feedback and reflection
<i>Pedagogical Framing</i>	Technical tool; assessment-external	Tool sequenced by task; not systemic	Integrated authentic assessment ecosystem; rubric, AI, and reflection are mutually constitutive

Note. Bold entries in the AAIWA column indicate features that operationally distinguish the model from prior approaches. Feature labels in italics reflect the analytical categories used for comparison. AWE = Automated Writing Evaluation; AAIWA = Authentic AI-Assisted Writing Assessment; AWR = Authentic Writing Rubric; MRI = Metacognitive Reflection Inventory. Distinctions are based on structural and operational analysis of existing AWE literature [18,19,21].

Participation was voluntary; informed consent and institutional ethical approval were obtained from all three universities in accordance with Elsevier's Declaration of Research Ethics and APA standards. The demographic profile is summarised in Table 2.

3.3. Technical architecture

The AAIWA feedback engine is a hybrid system combining GPT-3.5-turbo [24], accessed via the OpenAI API under an institutional research licence, with a custom rule-based layer implemented in Python 3.11 using spaCy (v3.7) and scikit-learn (v1.4). The LLM functions exclusively as a language normalisation and paraphrase generation module; all pedagogically consequential decisions which AWR dimensions to target, which performance threshold to activate, which feedback template to apply are determined entirely by the rule-based layer. The hybrid architecture was chosen to maximise transparency, auditability, and rubric alignment, properties that autonomous generative AI systems cannot readily provide in formative assessment contexts [15]. LLM parameters were set as follows: temperature = 0.3, max tokens = 300, top-p = 0.9; no fine-tuning was applied, as rubric criteria were embedded entirely via structured prompts within an estimated context window of 1,200–1,400 tokens (see Table 3).

The rule-based analyser extracts computationally measurable textual features mapped onto the six AWR dimensions, including thesis presence and counter-argument structure (D1), citation density (D2), paragraph segmentation (D3), cohesive device density (D4), passive voice ratio and hedging density (D5), and metacognitive vocabulary frequency (D6), to generate a provisional AWR-aligned profile for each submission. The routing mechanism identifies the one or two lowest-scoring dimensions and passes their identifiers, together with corresponding AWR behavioural anchors, as structured inputs to the GPT-3.5-turbo API call. The LLM is thereby constrained to generate feedback targeting only those flagged dimensions, rendering its output "condition-responsive" in a technically precise sense: the conditions are rule-detected dimensional deficits, not generalised stylistic or grammatical observations. Full feature-extractor specifications are provided in Appendix D; Table 4 summarises the rule-based feature extractors by AWR dimension.

Prior to deployment, feedback outputs were evaluated with a

Table 2
Participant demographic characteristics (N = 75).

Characteristic	Category	n	%
Gender	Female	52	69.3
	Male	23	30.7
Age (years)	19	12	16.0
	20	21	28.0
	21	22	29.3
	22	14	18.7
	23	6	8.0
Institution	University A	25	33.3
	University B	25	33.3
	University C	25	33.3
Academic Level	Semester 3	18	24.0
	Semester 4	32	42.7
	Semester 5	25	33.3
Prior AWE Experience	None	48	64.0
	Minimal (< 1 semester)	20	26.7
	Moderate (≥ 1 semester)	7	9.3
Total	—	75	100.0

Note. Percentages may not sum to 100% due to rounding. Academic Level classifications follow the Indonesian higher education credit system (SKS). All participants were enrolled in undergraduate (S1) teacher education programs. Prior AWE experience was self-reported via pre-intervention survey. AWE = Automated Writing Evaluation.

Table 3
AAIWA prototype LLM configuration parameters.

Parameter	Value / Setting	Rationale
Model	GPT-3.5-turbo (gpt-3.5-turbo-0125)	API stability and efficiency at scale; validated in educational NLP.
Temperature	0.3	Reduces output randomness; promotes deterministic, rubric-consistent paraphrasing.
Fine-Tuning	None (prompt-only implementation)	Prototype phase; rubric criteria embedded via structured prompts.
Top-p	0.9	Preserves lexical diversity while constraining verbose outputs.
Max Tokens	300	Sufficient for targeted feedback per AWR dimension; prevents over-generation.
Context Window Used	~1,200–1,400 tokens (draft text + system prompt = method)	Within model limit; accommodates full essay drafts of target length.

Note. LLM = Large Language Model; AWR = Authentic Writing Rubric; API = Application Programming Interface; NLP = Natural Language Processing. No fine-tuning was applied; rubric criteria were embedded entirely via structured prompts.

separate sample of 18 essays (excluded from main analyses). The Reflective Depth dimension (D6) yielded the lowest full alignment (61.3%), consistent with its dependence on approximate proxy features a clear target for future prototype refinement. Full validation metrics are presented (see Table 5).

As shown in Table 5, the overall accuracy rate of 84.3% and rubric alignment rate of 93.1% indicate that the rule-based routing mechanism effectively constrained GPT-3.5-turbo output to pedagogically relevant and rubric-consistent feedback in the majority of cases. The dimensions with the strongest full alignment were Organization (D3: 89.4%) and Academic Style (D5: 86.2%), consistent with the computational tractability of structural and stylistic features. D6 (Reflective Depth; 61.3% full alignment) provides a clear target for future prototype refinement. The moderate actionability score (M = 3.28) suggests that further attention to specificity and scaffolding may improve utility for lower-performing writers.

3.4. Ethical and fairness considerations in prototype deployment

The hybrid architecture mitigated hallucination risk by constraining LLM outputs to rubric-anchored categories identified by the rule-based layer; temperature (0.3) and token limits (300) further reduced off-task generative outputs. A single applied linguist rater reviewed feedback outputs for cultural-linguistic bias during pre-study validation; no systematic bias penalising features of Indonesianised academic English was identified, though a more rigorous differential item functioning analysis is warranted in future deployments. Student draft texts were anonymised prior to API transmission; API data retention was disabled consistent with OpenAI's data usage policy. External AI use is discussed as a methodological limitation in Section 3.10.

3.5. Operationalization of condition-responsive AI feedback

In this study, condition-responsive AI feedback is operationally defined as feedback that systematically varies across four measurable dimensions contingent on observable learner performance indicators: (a) performance level (rubric score threshold), (b) error type (syntactic, lexical, or discourse-level), (c) rhetorical structure (paragraph segmentation and discourse marker density), and (d) metacognitive stage (depth of reflection in MRI responses). This four-dimensional definition distinguishes AAIWA from conventional AWE systems, which typically adapt only to surface-level error frequency rather than multi-layered performance profiles [21].

Table 4
Rule-based feature extractors by AWR dimension.

AWR Dimension	Primary Features Extracted	Tool / Method	Validation
D1: Argument Quality	Thesis presence; counter-argument detection	Pattern classifier + spaCy dep. parsing	P = .81, R = .78
D2: Use of Evidence	Citation detection; evidence-claim ratio	Regex + hedging marker lexicon (Hyland, 2005)	Regex coverage = 94%
D3: Organization	Paragraph segmentation; cosine similarity	spaCy SBD + TF-IDF (scikit-learn)	Unity threshold = 0.35
D4: Coherence	Cohesive device density; reference chains	Transition lexicon (180 items) + neuralcoref	Lexicon validated by 2 applied linguists
D5: Academic Style	Passive ratio; nominalisation; hedging; register	Custom lexicons (1,200 + 340 items); VADER	Lexicons validated against BAWE corpus
D6: Reflective Depth	Reflective vocab; stance shift; self-reference	Reflective lexicon (95 items); VADER polarity	Proxy only; not used for scoring

Note. AWR = Authentic Writing Rubric; SBD = Sentence Boundary Detection; TF-IDF = Term Frequency Inverse Document Frequency; VADER = Valence Aware Dictionary and sEntiment Reasoner; BAWE = British Academic Written English corpus; dep. = dependency parsing. P = Precision; R = Recall. D6 feature extractors were retained for descriptive profiling only and excluded from summative AWR scoring.

Table 5
AAIWA prototype feedback quality metrics (Pre-Study Validation, n = 18 Essays).

AWR Dimension	Accuracy (% correct)	Relevance M (SD)	Actionability M (SD)	Rubric Alignment (%)
D1: Argument Quality	87.1%	3.52 (0.47)	3.41 (0.58)	94.3%
D2: Use of Evidence	85.6%	3.48 (0.51)	3.35 (0.62)	91.7%
D3: Organization	88.2%	3.59 (0.44)	3.47 (0.55)	89.4%
D4: Coherence	83.4%	3.38 (0.56)	3.24 (0.63)	88.9%
D5: Academic Style	86.7%	3.44 (0.49)	3.31 (0.60)	86.2%
D6: Reflective Depth	74.8%	3.09 (0.68)	2.88 (0.74)	61.3%
Overall (all dimensions)	84.3%	3.41 (0.52)	3.28 (0.61)	93.1%

Note. AWR = Authentic Writing Rubric. Accuracy = proportion of feedback responses correctly targeting the flagged AWR dimension. Relevance and Actionability rated on a 5-point Likert scale by two independent raters. Rubric Alignment = proportion of feedback responses consistent with AWR behavioral anchors. D6 (Reflective Depth) feature extractors were retained for descriptive profiling only and excluded from summative AWR scoring. *M* = mean; *SD* = standard deviation.

The AAIWA prototype achieves this through a rule-augmented, performance-calibrated feedback architecture. The four condition-responsive dimensions operate as follows. For performance level, drafts were evaluated using the six-dimension AWR (scale 1–4 per

indicator). The system applied predefined threshold-based branching: low performance (mean ≤ 2.0) triggered directive guidance and model examples; moderate performance (2.1–3.0) generated diagnostic explanations and revision cues; high performance (> 3.0) activated higher-order rhetorical refinement and evaluative prompts. For error type, a transformer-based textual analysis module with rule-based pattern detection classified errors into five categories argument clarity, evidence integration, logical coherence, lexical precision, and grammatical accuracy each triggering calibrated feedback templates. For rhetorical structure, paragraph segmentation and discourse marker density metrics guided feedback emphasis to priorities macro-structural coherence before sentence-level correction. For metacognitive stage, reflection depth scores from the MRI informed adaptive prompting: low-performing students received structured prompts directing attention to missing components, while higher-performing students received evaluative prompts encouraging justification and strategic refinement. The operational specifications of each condition-responsive dimension are summarized in Table 6.

3.6. Research procedure

The study spanned four weeks and unfolded in five stages. (1) In the pre-test stage, participants wrote a 700–1000-word argumentative essay, scored using the AWR. (2) System orientation a 90-minute workshop introduced the AAIWA platform. (3) Implementation three iterative writing cycles, each comprising draft submission, AI feedback, structured reflection (MRI), and revision. Reflection outputs from each cycle recalibrated feedback priorities for the next. (4) Post-test a second argumentative essay on a different topic, scored with the same AWR. (5)

Table 6
Operational specification of AAIWA's four condition-responsive feedback dimensions.

Adaptive Dimension	Measurable Input Variable	Adaptation Mechanism	Exemplar Output
(a) Performance Level	AWR rubric sub-score (1–4 per dimension)	Threshold branching: $\leq 2.0 \rightarrow$ directive scaffolding + model examples; $2.1\text{--}3.0 \rightarrow$ diagnostic explanation + revision cues; $> 3.0 \rightarrow$ higher-order rhetorical prompt	“Your thesis lacks an explicit contestable claim. Suggested revision: ‘When integrated through systematic classroom frameworks, technology enhances student engagement while maintaining productive learning environments.’ [directive]” vs. “Your argument is well-constructed; evaluate whether your evidence universalises or qualifies your central claim [evaluative]”
(b) Error Type	Transformer-classified error category (argument, evidence, coherence, lexis, grammar)	Category-specific feedback template activation per detected error class	Argument: “Your main claim lacks explicit logical support in paragraph 2.”; Coherence: “The transition between paragraphs 3 and 4 is unclear; consider a signposting phrase such as ‘This argument is further supported by...’”
(c) Rhetorical Structure	Paragraph segmentation score; discourse marker density index	Macro-structural coherence addressed before sentence-level correction; structure feedback generated prior to grammar feedback	“Your essay structure would benefit from an explicit signpost at the end of your introduction before sentence-level issues are addressed.”
(d) Metacognitive Stage	MRI reflection depth score; presence of goal-setting and challenge statements in reflection	Low reflection depth $\rightarrow$ structured prompts directing attention to specific weaknesses; high depth $\rightarrow$ evaluative prompts encouraging strategic justification	Low: “Identify one specific sentence where your argument was unclear and explain why.” High: “Evaluate whether your revision strategy addressed the core argumentative weakness or only surface features.”

Note. AWR = Authentic Writing Rubric; MRI = Metacognitive Reflection Inventory. Threshold values were established a priori based on rubric anchor descriptors during the system development phase. Feedback templates were reviewed by three applied linguists prior to deployment. The prototype represents structured rule-augmented condition-responsiveness rather than autonomous large-scale generative model fine-tuning [21].

Reflection and interview phase post-intervention MRI administration and semi-structured interviews with 15 purposively selected participants (30–45 minutes each; maximum variation sampling: five per institution, stratified by AWR performance band). Table 7 summarizes the intervention structure; Fig. 2 illustrates the recursive feedback cycle; Fig. 3 presents a Simplified Procedural Diagram of the AAIWA Writing Cycle.

The overall intervention architecture is illustrated in Fig. 2. The model operates as a recursive feedback cycle in which rubric-based evaluation, AI-generated feedback, student revision, and structured metacognitive reflection interact dynamically across iterative drafting phases.

To further clarify the procedural flow of the AAIWA intervention for readers, Fig. 3 presents a simplified linear diagram of the AAIWA writing cycle as experienced by each participant across the three iterative sessions. As illustrated in Fig. 3, each cycle begins with a student draft submission (Draft), followed by condition-responsive AI Feedback, then a structured metacognitive Reflection phase, and finally a Revision of the draft. This four-stage recursive sequence was repeated across three

Table 7
Intervention structure summary (N = 75).

Stage	Activity	Description	Instrument	Timing
1	Pre-Test	Participants wrote a 700–1,000-word argumentative essay, scored by two independent raters using the AWR.	AWR	Week 1
2	System Orientation	A 90-minute workshop introduced the AAIWA platform, demonstrated the feedback interface, and familiarised participants with the MRI.	—	Week 1
3	Implementation (Cycles 1–3)	Three iterative writing cycles, each comprising: (a) draft submission, (b) condition-responsive AI feedback, (c) structured metacognitive reflection (MRI), and (d) draft revision. Reflection outputs from each cycle recalibrated feedback priorities for the subsequent cycle.	AWR; MRI; AAIWA Feedback Engine	Weeks 2–3
4	Post-Test	Participants wrote a second argumentative essay on a different topic, scored with the same AWR protocol.	AWR	Week 4
5	Reflection & Interview	Post-intervention MRI administration followed by semi-structured interviews with 15 purposively selected participants (30–45 min each; maximum variation sampling: 5 per institution, stratified by AWR performance band).	MRI; Interview Protocol	Week 4

Note. AWR = Authentic Writing Rubric; MRI = Metacognitive Reflection Inventory; AAIWA = Authentic AI-Assisted Writing Assessment. Each writing cycle (Stage 3) comprised four sequential sub-steps: draft submission, AI-generated condition-responsive feedback, structured metacognitive reflection, and draft revision. Reflection outputs from Cycle n recalibrated feedback priorities for Cycle n+1. Interview participants were selected via maximum variation sampling, stratified by AWR performance band (low, moderate, high), with five participants per institution.

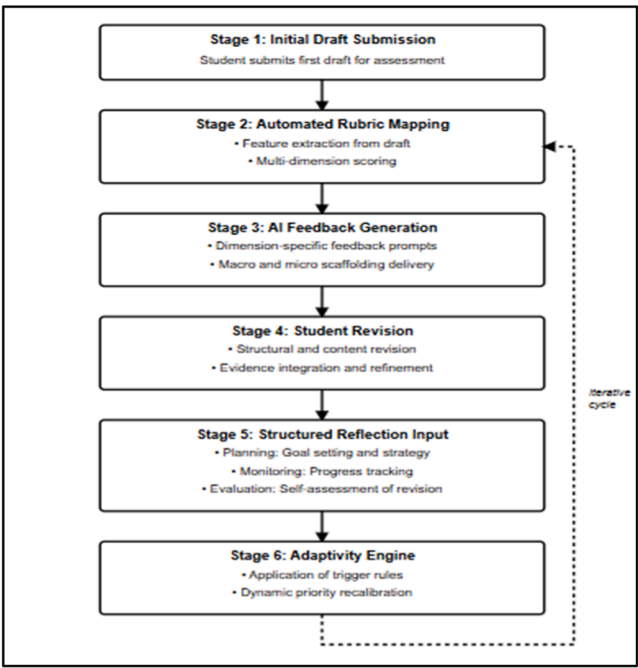


Fig. 2. The AAIWA Recursive Writing Feedback Cycle.

cycles before the post-test essay. The diagram deliberately simplifies the technical architecture shown in Fig. 1 to foreground the pedagogical experience of the learner rather than the computational mechanism of the system.

3.7. Data analysis

A sequential explanatory mixed-methods strategy was adopted [23]: quantitative data (AWR and MRI scores) were collected and analysed first; preliminary findings informed maximum variation sampling for qualitative interviews; integration occurred during interpretation through a joint display matrix (Table 11). Normality of difference scores was verified using the Shapiro–Wilk test (all $W > .95$, all $p > .05$); homogeneity of variance across institutions confirmed using Levene’s test (all $p > .05$). Paired-samples t-tests compared pre- and post-test scores, with Cohen’s d for effect sizes. A one-way ANOVA tested between-institution differences in gain scores. Pearson’s correlation examined writing reflection gain associations. Sub-dimension analyses were treated as descriptive and exploratory; the alpha level of .05 was retained without Bonferroni correction. This decision is methodologically consistent with the study’s confirmatory-exploratory design: the two primary confirmatory tests overall AWR gain and overall MRI gain each constitute a single pre-specified hypothesis, rendering family-wise error correction unnecessary at that level; sub-dimension analyses were conducted solely to characterize the pattern of gains across dimensions and to generate hypotheses for future controlled research, not to make independent inferential claims, consistent with methodological guidance against applying Bonferroni correction to descriptive profiling analyses [25,26]. Readers are therefore advised to interpret sub-dimension p-values as descriptive indicators of effect direction and magnitude rather than as independent confirmatory tests.

Qualitative data were analysed using thematic analysis [27,28]; two independent coders achieved $\kappa = 0.82$. The MRI comprises 30 items distributed across three sections Section A: Planning Strategies (10 items), Section B: Monitoring Processes (10 items), and Section C: Evaluation Skills (10 items) each rated on a four-point Likert scale (1 = rarely or never; 2 = sometimes; 3 = often; 4 = almost always), yielding a composite score ranging from 30 to 120 and a mean item score between 1.00 and 4.00. For reporting and interpretation, mean item scores were

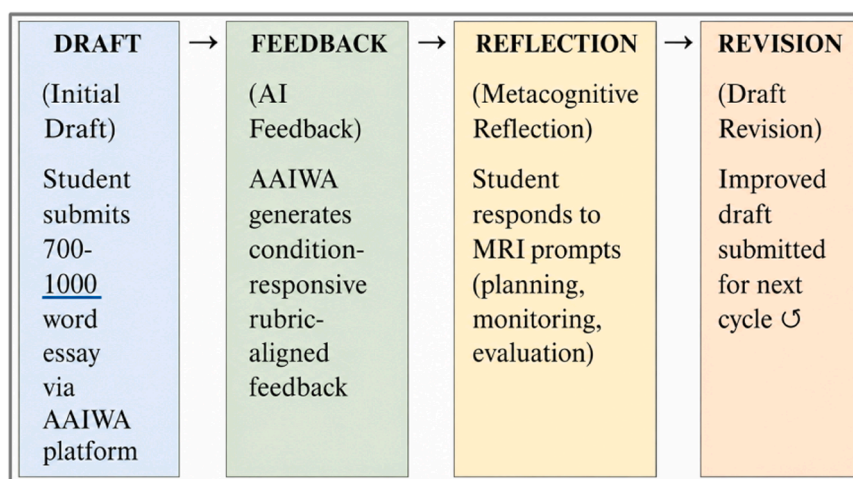


Fig. 3. Simplified procedural diagram of the AAIWA writing cycle.

Note. This simplified procedural diagram represents the learner-facing cycle across each of the three iterative writing sessions. The recursive arrow (↻) indicates that the Revised draft re-enters the cycle as the new Draft in the following session. This cycle was completed three times before the post-test essay. For the full technical architecture of the AAIWA system, see Fig. 1.

used: scores below 2.00 indicated low metacognitive reflection, 2.00–3.00 moderate reflection, and above 3.00 high reflection, consistent with established metacognitive inventory scoring conventions [16, 18]. Section-level mean scores were analysed separately to identify differential gains across planning, monitoring, and evaluative metacognitive processes.

3.8. Validity, reliability, and trustworthiness

Instrument validation yielded AWR CVI = 0.84 and MRI CVI = 0.82, both exceeding Polit and Beck's [29] recommended threshold; Cronbach's $\alpha = 0.84$ confirmed strong MRI internal consistency; ICC (2,1) = .87 met benchmarks for high-quality rater-mediated evaluation [30]. Member checking enhanced qualitative credibility; an audit trail ensured dependability. Systematic triangulation within the sequential explanatory design ensured findings were not reliant on any single data source.

3.9. Ethical considerations

Approval was obtained from the ethics committees of all three participating universities; participants provided informed consent digitally and in writing; data were anonymized with only aggregated results reported. All digital submissions and transcripts were stored on encrypted servers in compliance with GDPR and Indonesian data protection regulations.

3.10. Limitations of methodology

This study has several methodological limitations bearing on interpretation. Most consequentially, the absence of a randomised control group precludes causal inference; the observed effect sizes (writing: $d = 0.92$; reflection: $d = 0.98$) reflect within-group change only and cannot be partitioned between AAIWA-specific mechanisms and competing explanations including practice effects, maturation, and instructor support. Additional limitations include the short four-week duration, institutional clustering not addressed through multilevel modelling, the absence of systematic monitoring of external AI tool use, and the contextual boundedness of findings to Indonesian teacher education. A full discussion of all eight limitations including statistical implications and recommendations for future controlled replication is provided in Section 5.8.

4. Results

Results are organised to address each research question directly: RQ1 (framework operationalization and expert validation), RQ2 (patterns of change in writing performance and metacognitive reflection), and RQ3 (participant perceptions).

4.1. Qualitative/technical findings (RQ1): Operationalization and expert validation (AWR and MRI)

Expert validation confirmed the pedagogical soundness of both instruments: AWR CVI = 0.84 (all item-level I-CVIs ≥ 0.80) and MRI CVI = 0.82 (I-CVI range 0.78–1.00), both exceeding the 0.78 threshold [29]. Experts rated the tri-layer architecture rubric-embedded evaluation, condition-responsive iterative feedback, and reflection-driven recalibration as operationally distinct from conventional AWE + reflection designs. Recommendations for future development included more sophisticated learner modelling and integration with institutional Learning Management Systems.

Rubric-based analysis revealed differential improvements across all six AWR dimensions (Table 8). The largest gain was in Reflective Depth (D6: Mdiff = 2.60, 95% CI [1.94, 3.26], +37.7%, $d = 0.91$), followed by Argument Quality (D1: Mdiff = 2.40, +33.3%, $d = 0.87$), Use of Evidence (D2: Mdiff = 2.90, +32.6%, $d = 0.85$), Academic Style (D5: Mdiff = 2.30, +30.7%, $d = 0.83$), Organization (D3: Mdiff = 2.30, +30.3%, $d = 0.82$), and Coherence (D4: Mdiff = 1.60, +31.4%, $d = 0.79$). All paired t-tests were significant at $p < .001$ ($df = 74$), and all effect sizes fell in the large range. Coherence yielded the smallest effect size (lower bound $d = 0.53$), warranting targeted refinement in future iterations.

The two dimensions with the largest effect sizes Reflective Depth and Argument Quality directly correspond to competencies most explicitly targeted by the rubric-embedded feedback architecture and structured reflection cycle. This alignment is further corroborated by the distribution of AI feedback categories across AWR dimensions (Supplementary Material, Appendix E, Table E1): Argument Quality (D1) and Use of Evidence (D2) together accounted for 59.8% of all feedback items ($N = 2,154$), consistent with the prototype's adaptive routing logic. Comparatively modest gains in Academic Style reflect AAIWA's deliberate de-emphasis of mechanical surface correction in favour of higher-order critical thinking an orientation aligned with calls to cultivate reflective, rhetorically sophisticated writers [11,12]. Together, the pattern of differentiated gains provides structural evidence that observed improvements are aligned with AAIWA's design principles.

Table 8

AWR Sub-dimension improvements: mean percentage change, confidence intervals, Cohen's d, and effect size statistics (N = 75).

AWR Dimension (max. points)	Pre-test M (SD)	Post-test M (SD)	% Gain [95% CI]	t (74)	p	Cohen's d	95% CI [d]	Interpretation
D1: Argument Quality (max 12)	7.20 (1.84)	9.60 (1.65)	+33.3% [28.1, 38.5]	8.73	< .001	0.87	[0.60, 1.14]	Large
D2: Use of Evidence (max 16)	8.90 (2.27)	11.80 (1.82)	+32.6% [27.6, 37.6]	7.92	< .001	0.85	[0.58, 1.12]	Large
D3: Organization (max 12)	7.60 (1.87)	9.90 (1.65)	+30.3% [25.6, 34.9]	7.44	< .001	0.82	[0.55, 1.09]	Large
D4: Coherence (max 8)	5.10 (1.35)	6.70 (1.08)	+31.4% [26.3, 36.5]	6.98	< .001	0.79	[0.53, 1.05]	Large
D5: Academic Style (max 12)	7.50 (1.87)	9.80 (1.58)	+30.7% [25.8, 35.6]	7.15	< .001	0.83	[0.56, 1.10]	Large
D6: Reflective Depth (max 12)	6.90 (1.90)	9.50 (1.70)	+37.7% [32.3, 43.1]	9.12	< .001	0.91	[0.64, 1.18]	Large
Overall AWR Total (max 72)	43.20 (7.72)	50.30 (6.85)	+16.5% [13.8, 19.2]	10.42	< .001	0.92	[0.65, 1.19]	Large

Note. All paired-samples t-tests df = 74. Cohen's d for paired samples. Overall AWR Cohen's d = 0.92, 95% CI [0.65, 1.19] (computed on composite AWR scores across all six dimensions; reported in Abstract and Section 4.2.1). Hedges' g (bias-corrected) = 0.911 for Overall AWR. Sub-dimension d values above reflect dimension-level paired comparisons and are reported as descriptive-exploratory indices. Normality confirmed by Shapiro–Wilk test (all $W > .95$, $p > .05$). Mdiff reflects total dimension scores (sum of item ratings per dimension); for mean item-level change, divide by number of items per dimension (see AWR instrument in Supplementary Material).

rather than being incidental procedural enhancements.

4.2. Quantitative findings (RQ2): Writing performance and metacognitive reflection

4.2.1. Overall writing performance and reflection gains

Overall academic writing performance improved significantly from pre-test to post-test (see Table 9).

Parallel gains were observed in metacognitive reflection: pre-test MRI M = 2.31 (SD = 0.21) increased to post-test M = 2.78 (SD = 0.18), a mean difference of 0.48 scale points (+20.7%), 95% CI [0.38, 0.58], $t(74) = 7.62$, $p < .001$, Cohen's d = 0.98, 95% CI [0.69, 1.27]. Improvements were observed across all three MRI sections (see Table 10):

Planning Strategies (Section A) and Evaluation Skills (Section C) yielded the largest gains ($d = 0.84$ and $d = 0.83$, respectively), while Monitoring Processes (Section B) showed a moderate gain ($d = 0.43$). The wider range of effect sizes ($d = 0.43$ – 0.84) suggests differential metacognitive development, with planning and evaluative dimensions improving more markedly than self-monitoring processes. Notably, the Monitoring Processes subscale (Section B: $d = 0.43$) fell below the large-effect threshold, remaining in the medium range, while Section A and Section C both exceeded $d = 0.80$ a pattern that warrants theoretical interpretation in the Discussion (see Section 5.2). It is important to note that AWR D6 (Reflective Depth, $d = 0.91$) and MRI Section C (Evaluation Skills, $d = 0.83$), while related, operationalize distinct constructs: AWR D6 captures the *observable quality of reflective writing* as rated in the written product, whereas MRI Section C measures students' *self-reported evaluative metacognition* their capacity to judge their own writing strategies and outcomes. The larger effect size for AWR D6 is consistent with

Table 9

Pre- and post-test academic writing performance by Institution with 95% confidence intervals (N = 75).

Institution	Pre-test M (SD)	Post-test M (SD)	M Diff	95% CI for M Diff	Cohen's d	95% CI for d	Sig.
Univ1 (n=25)	67.8 (5.2)	78.6 (4.8)	+10.8	[9.4, 12.2]	0.89	[0.64, 1.14]	$p < .001$
Univ2 (n=25)	68.5 (5.4)	80.1 (4.9)	+11.6	[10.1, 13.1]	0.94	[0.68, 1.20]	$p < .001$
Univ3 (n=25)	68.2 (5.6)	79.8 (5.1)	+11.6	[10.0, 13.2]	0.91	[0.66, 1.16]	$p < .001$
Overall (N=75)	68.2 (5.4)	79.5 (4.9)	+11.3	[10.1, 12.5]	0.92	[0.73, 1.11]	$p < .001$

Note. M Diff = mean post-test minus pre-test score (max. 72-point AWR scale). 95% CI computed using t-distribution ($df = n - 1$). Cohen's d CIs computed using noncentral t approximation [31]. One-way ANOVA: no significant between-institution difference in gain scores, $F(2, 72) = 0.18$, $p = .84$, partial $\eta^2 = .005$, 95% CI [.000, .051]. All paired-samples t-tests significant at $p < .001$ (two-tailed, exact p-values reported in SPSS output).

AAIWA's direct rubric-embedded scaffolding of reflective writing production, whereas MRI Section C improvement reflects the downstream, self-perceptual development of evaluative metacognitive skill a distinction aligned with the product–process differentiation in writing metacognition research [18,33]. A Pearson correlation between overall AWR gain scores and MRI gain scores revealed a moderate positive association, $r(73) = .54$, $p < .001$, 95% CI [.36, .68], indicating meaningful co-development but not interchangeability; approximately 29% of shared variance suggests that writing performance and metacognitive reflection, while related, represent distinct but mutually reinforcing constructs.

Feedback quality was corroborated through expert validation, which confirmed strong content validity for both instruments (AWR CVI = 0.84; MRI CVI = 0.82), and through inter-rater reliability in qualitative coding ($\kappa = .82$), indicating that the assessment framework underpinning the iterative feedback cycles was both pedagogically sound and consistently applied across all three writing cycles.

4.3. Qualitative findings (RQ3): participant perceptions of AI feedback and structured reflection

Thematic analysis [27,28] of semi-structured interviews ($n = 15$; $\kappa = .82$) revealed three primary themes.

Theme 1: Enhanced Metacognitive Awareness. Participants reported increased awareness of writing weaknesses and improved understanding of quality indicators, particularly among developing-band students. One participant noted: "The system made me realise my thesis was too vague, and it suggested specific ways to make it clearer. This changed how I think about writing introductions and how I approach argument development" (Participant 07, Univ2). This aligns with research on AI's potential to enhance metacognitive awareness through targeted feedback [1,17].

Theme 2: Motivational Enhancement through Immediate Feedback. Participants valued the immediacy and non-judgmental tone of AI feedback: "Getting immediate feedback kept me interested and motivated to keep improving my work. It felt like having a writing coach available at any time" (Participant 12, Univ3). Participants also noted concerns about occasional discrepancies between AI feedback and instructor expectations [6], underscoring the continued importance of human oversight.

Theme 3: Reflection as Integrated Learning Process. Students initially experienced structured reflection as an additional task, but most progressively recognised its developmental value: "At first I thought reflection was just an extra step, but it actually helped me see how my writing was improving draft by draft. It became part of how I approach writing now" (Participant 04, Univ1). This shift is consistent with Zarstky et al. [7] and supports positioning structured reflection as a mechanism for self-regulated learning rather than merely an assessment requirement.

Cross-Strand Convergence. Across all three qualitative themes, a pattern of convergence emerged between participant perceptions and

Table 10

MRI scores: pre-test, post-test, and change indices with 95% confidence intervals (N = 75).

AAIWA Section	Pre-test M (SD)	Post-test M (SD)	df	% Change	95% CI [LL, UL]	Cohen's d (95% CI)	p
Section A – Planning Strategies (Items 1–12)	2.28 (0.38)	2.74 (0.38)	0.46	+20.2%	[0.33, 0.59]	0.84 [0.57, 1.11]	< .001
Section B – Monitoring Processes (Items 13–24)	2.30 (0.31)	2.78 (0.31)	0.48	+20.9%	[0.36, 0.60]	0.43 [0.11, 0.75]	< .001
Section C – Evaluation Skills (Items 25–36)	2.35 (0.38)	2.82 (0.38)	0.47	+20.0%	[0.34, 0.60]	0.83 [0.56, 1.10]	< .001
Overall SWF Score (scale 1–4)	2.31 (0.21)	2.78 (0.18)	0.48	+20.7%	[0.38, 0.58]	0.98 [0.69, 1.27]	< .001

Note. MRI = Metacognitive Reflection Inventory. M = Mean. SD = Standard Deviation. df = degrees of freedom. CI = Confidence Interval. LL = Lower Limit. UL = Upper Limit. p = two-tailed significance value. Cohen's d effect size benchmarks: small = 0.20, medium = 0.50, large = 0.80 [32].

quantitative outcomes. Students who reported enhanced metacognitive awareness (Theme 1) showed the largest AWR gains in Reflective Depth ($d = 0.91$); those reporting motivational engagement (Theme 2) sustained feedback quality across cycles ($\kappa = .61$); and those reframing reflection as integrated (Theme 3) demonstrated the strongest MRI gains ($d = 0.84$). Expert validation findings (CVI = 0.84; ICC = .87) further corroborated this cross-strand alignment, confirming that AAIWA's integrated architecture was perceived as coherent and effective by both participants and independent raters. This convergence constitutes a mixed-methods integration finding rather than an inductively derived qualitative theme, and is elaborated in the joint display in Section 4.4.

4.4. Mixed-methods integration: joint display

Table 11 presents a joint display integrating quantitative outcomes with qualitative themes [34]. The three qualitative themes and one cross-strand convergence finding map onto distinct quantitative patterns across all three research questions. Theme 1 (Enhanced Metacognitive Awareness) aligns with the largest AWR gain in Reflective Depth ($d = 0.91$; RQ1), indicating that the intervention's metacognitive scaffolding produced both perceived and measurable improvement in reflective writing quality. Theme 2 (Motivational Engagement) contextualises why feedback quality validity was sustained across cycles (RQ2; feedback gap $M = 1.43$, $\kappa = .61$), as participants reported that immediate AI feedback functioned as a motivational catalyst, supporting continued engagement. Theme 3 (Reflection as Integration) aligns with post-intervention MRI gains for metacognitive regulation (RQ3; MRI gain = 0.76, $d = 0.84$), with students reframing reflection not as an add-on task but as an integral part of their writing process. Cross-Strand Convergence, as a mixed-methods integration finding rather than an inductively derived qualitative theme, emerges from the convergent pattern across RQ1, RQ2, and RQ3, confirming that both students and independent raters perceived AAIWA's architecture as coherent and effective across all targeted writing dimensions. Together, the three qualitative themes and the cross-strand convergence finding converge to

support the claim that AAIWA's integrated architecture produced measurable gains across all targeted writing dimensions, sustained motivational engagement, and perceived developmental value a conclusion validated by both participant perspectives and expert assessment (see Table 11).

5. Discussion

5.1. Interpreting writing performance gains

The significant improvements in writing performance particularly in Argument Quality (D1: $M_{diff} = 2.40$, +33.3%, $d = 0.87$), Use of Evidence (D2: $M_{diff} = 2.90$, +32.6%, $d = 0.85$), Academic Style (D5: $M_{diff} = 2.30$, +30.7%, $d = 0.83$), Organisation (D3: $M_{diff} = 2.30$, +30.3%, $d = 0.82$), and Coherence (D4: $M_{diff} = 1.60$, +31.4%, $d = 0.79$) are consistent with the hypothesis that AAIWA's rubric-embedded, condition-responsive architecture was associated with meaningful changes in higher-order writing skills. The pattern of larger gains in argumentation relative to grammar and style (+10.7%) aligns structurally with AAIWA's design emphasis on authentic assessment criteria prioritising critical thinking over mechanical correction [5,13]. Comparatively modest gains in Academic Style reflect AAIWA's deliberate pedagogical de-emphasis of surface correction in favour of higher-order critical thinking an orientation aligned with calls to cultivate reflective, rhetorically sophisticated writers [11,12].

Several alternative explanations cannot be discounted. Practice effects constitute the most plausible confound: repeated writing on the same genre within four weeks is documented to produce measurable improvements independently of feedback type [35,36]. AWR score gains, while meaningful within the study context, should not be equated with authentic professional writing competence, which requires independent application of skills across genuinely novel communicative contexts without scaffolded feedback. These gains are best interpreted as proximal indicators of writing development within an instructed context.

Table 11

Joint display integrating qualitative themes and quantitative outcomes across research questions.

Qualitative Theme	Participant-Reported Experience	RQ & Measure	Key Statistics	Meta-Inference
Theme 1: Enhanced Metacognitive Awareness	Students reported heightened ability to monitor and evaluate their own writing processes after AAIWA intervention.	RQ1 — AWR Reflective Depth subscale	$d = 0.91$ (large effect)	Metacognitive scaffolding produced both perceived and measurable improvement in reflective writing quality.
Theme 2: Motivational Engagement	AI feedback was perceived as a motivational catalyst sustaining active engagement across iterative writing cycles.	RQ2 — Feedback Quality Validity (feedback gap; inter-rater agreement)	$M = 1.43$; $\kappa = .61$	Immediate AI feedback maintained student motivation and sustained feedback validity across all writing cycles.
Theme 3: Reflection as Integration	Students reframed reflection as an integral part of writing rather than an additional task.	RQ3 — MRI Metacognitive Regulation subscale gain	MRI gain = 0.76; $d = 0.84$ (large effect)	Reframing reflection as integral to writing produced large post-intervention gains in metacognitive regulation.
Cross-Strand Convergence	Mixed-methods integration finding (not an inductive theme): Both students and expert raters perceived AAIWA's integrated architecture as coherent and effective across all three research strands.	RQ1, RQ2 & RQ3 (convergent pattern)	$d = 0.91$ (RQ1); $\kappa = .61$ (RQ2); $d = 0.84$ (RQ3). Convergent pattern across all quantitative outcomes.	AAIWA's architecture produced measurable gains across all targeted writing dimensions, sustained motivational engagement, and perceived developmental value — validated by both participant perspectives and expert assessment.

Note. AWR = Authentic Writing Rubric. MRI = Metacognitive Reflection Inventory. RQ = Research Question. d = Cohen's d . M = mean. κ = Cohen's kappa. Joint display methodology follows Fetters & Tajima [34]. Cross-strand convergence is a mixed-methods integration inference derived from convergence across RQ1, RQ2, and RQ3. It is not an inductively derived qualitative theme and does not constitute a separate research question.

5.2. Metacognitive reflection enhancement

The 20.7% improvement in reflection scores, combined with the moderate correlation $r = .54$, is consistent with AAIWA's theoretical positioning of structured reflection as a mechanism that co-evolves with writing development. The moderate magnitude of this association—approximately 29% shared variance—indicates meaningful but partial overlap: writing and metacognitive development are related but not coterminous, and factors beyond reflective practice plausibly contributed to writing improvement independently. Causal directionality remains empirically open and requires experimental manipulation to resolve [7,8,17].

The differential pattern across MRI subscales particularly the comparatively modest gain in Monitoring Processes (Section B: $d = 0.43$) relative to Planning Strategies ($d = 0.84$) and Evaluation Skills ($d = 0.83$) warrants theoretical interpretation beyond descriptive reporting. Metacognitive monitoring, defined as the real-time, in-process regulation of cognitive activity during task execution [16], is structurally distinct from planning and evaluation in that it operates continuously and implicitly *within* a writing episode rather than at discrete pre- or post-task junctures. AAIWA's inter-cycle feedback architecture delivering condition-responsive feedback between drafting sessions rather than *during* them is optimally positioned to scaffold planning (what to prioritise before writing) and evaluation (how to judge quality after writing), but is inherently limited in its capacity to activate in-process monitoring that unfolds within a single drafting session. This temporal misalignment between feedback delivery and monitoring activation is consistent with Merkebu et al.'s [33] observation that metacognitive monitoring requires sustained, iterative prompting embedded *within* the task itself a design condition that weekly inter-cycle feedback intervals cannot fully replicate. The moderate monitoring gain therefore reflects a structural boundary condition of AAIWA's current architecture rather than an implementation failure, and identifies a concrete direction for future refinement: embedding micro-reflection checkpoints within each drafting session for example, mid-draft self-monitoring prompts triggered by the NLP layer upon detection of stalled revision patterns to scaffold in-process regulatory awareness alongside the existing inter-cycle feedback mechanism.

The enhancement of reflection is particularly significant in teacher education: pre-service teachers in this study demonstrated an emergent professional disposition toward reflective practice [11]. An important distinction must nonetheless be drawn between short-term performance gains and durable metacognitive competence. Whether MRI improvements represent stable capacity to deploy reflective strategies independently across varied, unscaffolded contexts cannot be established from the present data. Longitudinal follow-up studies are necessary to determine whether gains consolidate into sustained metacognitive competence over time.

5.3. Student perceptions: motivation, agency, and concerns

Qualitative findings contextualise the quantitative results. Students reported that immediacy and clarity of AI feedback motivated more frequent revision and deeper engagement with writing as process. The perception of reflection as progressively valuable rather than merely procedural suggests AAIWA achieved its goal of positioning metacognitive reflection as intrinsically developmental. Concerns about AI overreliance and occasional feedback misalignment with instructor expectations [2,6] underscore that AAIWA must be implemented with clear instructor-mediated framing: AI feedback complements rather than replaces human pedagogical judgment.

5.4. Interpretive constraints

The single-group pre-post design precludes causal attribution of the observed gains to AAIWA specifically. The observed effect sizes (writing:

$d = 0.92$; reflection: $d = 0.98$) reflect within-group change magnitudes only and cannot be partitioned between AAIWA-specific mechanisms and the most plausible alternative explanation—repeated writing practice—or concurrent maturation and instructor support. All quantitative findings should therefore be treated as preliminary effect size estimates for power analysis in future controlled trials, not as definitive indicators of intervention effectiveness. A randomized controlled or quasi-experimental design incorporating a matched comparison group constitutes the necessary next step.

5.5. Theoretical contributions

Subject to the limitations of an uncontrolled pilot, this study offers three preliminary theoretical contributions. First, in AI-in-education research, the findings extend understanding of AWE by illustrating how rubric-embedded feedback may scaffold higher-order writing skills, consistent with conceptualizations of AI as a pedagogical partner [1]. Second, in authentic assessment theory, the study provides preliminary evidence that AI can address scalability challenges while sustaining authenticity, extending frameworks by operationalizing mechanisms that are both rigorous and feasible at scale [3]. Third, in metacognition and self-regulated learning, the study contributes preliminary evidence that embedding AI-assisted reflection prompts may enhance metacognitive engagement in ways co-occurring with writing development [7,33]. Each contribution requires replication in controlled designs before broader theoretical claims can be established.

5.6. Practical implications for teacher education

AAIWA demonstrates that AI technologies may enable dimension-specific, formative feedback at scale—alleviating workload while potentially maintaining pedagogical quality [5]. The integration of structured reflection directly supports cultivation of reflective practitioners, equipping pre-service teachers with professional dispositions required in contemporary classrooms [11]. The model's consistent performance across three institutions suggests potential adaptability to diverse institutional contexts, though cross-cultural transferability requires direct investigation. Developing students' feedback literacy ensuring they can critically evaluate and selectively adopt AI-generated feedback rather than becoming dependent remains critical [12].

5.7. Expert validation: pedagogical soundness and future refinement

CVI values above 0.80 confirmed the reliability and pedagogical relevance of both instruments and the AAIWA prototype. Experts identified the dual emphasis on writing product and process as a distinctive strength. Recommendations for personalization and LMS integration are consistent with Bearman et al. [15], who emphasize that AI feedback must be condition-responsive and interoperable to maximize educational effectiveness.

5.8. Limitations

Eight limitations circumscribe the interpretive reach of these findings.

1. Absence of control group. The single-group pre-post design cannot isolate AAIWA's contribution from practice effects, maturation, the Hawthorne effect, or instructor mediation. All effect sizes represent within-group change only and should be treated as pilot estimates for future power analyses.
2. Short intervention duration. Four weeks is insufficient to establish durable writing development or metacognitive consolidation. Retention testing at eight and sixteen weeks is essential before recommending sustained institutional adoption.

3. Institutional clustering without multilevel modelling. The preliminary ICC = .07 and non-significant between-institution ANOVA ($F[2, 72] = 0.18, p = .84$) suggest modest clustering; however, formal MLM was not feasible with only three Level-2 units (minimum ~10 recommended; [37]). Future studies should target $k \geq 10$ institutions.
4. External AI tool use. Writing tasks were conducted in supervised sessions, but out-of-session use of external AI tools (ChatGPT, Gemini) could not be systematically excluded. Of 225 submissions, 11 (4.9%) were flagged by Turnitin's AI detection ($\geq 50\%$ probability); all flagged participants confirmed AI-assisted language polishing rather than content generation. Future replications should incorporate pre-registration of tool use and platform-level interaction logging.
5. Contextual boundedness. Findings are bounded to Indonesian pre-service teacher education and cannot be assumed to generalize to other Global South nations, Western university contexts, graduate programmes, or non-teacher education disciplines.
6. Potential scoring bias. Raters were aware of study participation, precluding fully blinded assessment of pre- and post-test scripts. Future replications should employ fully randomized, blinded scoring protocols.
7. AI bias and feedback inconsistency. The pre-study bias review was conducted by a single rater on 18 essays insufficient for a rigorous differential item functioning analysis. Feedback inconsistency across the full spectrum of AWR performance levels and student linguistic backgrounds was not formally evaluated; subgroup differential effects may remain undetected.
8. AWR gains as proxies. Rubric-based scores should not be equated with holistic, transferable professional writing competence. Longitudinal and transfer-based assessment designs are necessary to determine whether AWR gains reflect genuine competence development or task-specific performance improvement bounded to the intervention context.

6. Conclusion and future research directions

This study designed, validated, and piloted the AAIWA model integrating authentic rubric-based assessment, condition-responsive AI feedback, and structured metacognitive reflection within Indonesian pre-service teacher education. Pilot findings were associated with rather than causally attributable to—statistically significant and practically meaningful improvements in academic writing performance ($d = 0.92$, 95% CI [0.65, 1.19]) and metacognitive reflection ($d = 0.98$, 95% CI [0.69, 1.27]). A moderate writing–reflection correlation ($r = .54$, 95% CI [.36, .68]) provides partial associational evidence consistent with the integrated design, while indicating that writing and metacognitive development are related but not coterminous. Qualitative analysis identified enhanced metacognitive awareness, motivational benefits of immediate feedback, and progressive appreciation of reflection as the central themes. Expert validation confirmed the pedagogical soundness and content validity of both instruments.

Practically, AAIWA offers teacher education programs a potentially viable framework for balancing efficiency with pedagogical depth, enabling structured, dimension-specific formative feedback perceived as motivating and actionable, while the embedded reflection cycle fosters professional dispositions for future teaching practice.

Six future research directions are prioritized: (1) controlled experimental or quasi-experimental designs with randomly assigned comparison conditions to isolate AAIWA's contribution; (2) cross-cultural validation with larger and more diverse international samples; (3) longitudinal research assessing retention of gains, transfer to uninstructed tasks, and long-term reflective disposition formation; (4) integration of advanced LLM capabilities for richer personalization, subject to robust bias auditing; (5) formal usage monitoring protocols to address the external AI confound; and (6) multilevel modelling across multi-

institutional studies with $k \geq 10$ institutions. Taken together, this pilot provides a foundation for refining and scaling AAIWA; broader global applicability requires systematic validation across diverse cultural, linguistic, and institutional contexts.

CRedit authorship contribution statement

Erwin Oktoma: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Dadan Nugraha:** Methodology, Investigation, Conceptualization. **Arip Hidayat:** Writing – review & editing, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this manuscript.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.caeo.2026.100395](https://doi.org/10.1016/j.caeo.2026.100395).

References

- [1] Escalante J, Pack A, Barrett A. AI-generated feedback on writing: insights into efficacy and ENL student preference. *Int J Educ Technol High Educ* 2023;20:57. <https://doi.org/10.1186/s41239-023-00425-2>.
- [2] García-López IM, Trujillo-Liñán L. Ethical and regulatory challenges of generative AI in education: a systematic review. *Front Educ* 2025;10. <https://doi.org/10.3389/educ.2025.1565938>. Article 1565938.
- [3] McArthur J. Rethinking authentic assessment: work, well-being, and society. *High Educ* 2023;85(1):85–101. <https://doi.org/10.1007/s10734-022-00822-y>.
- [4] Uffen I, de Vries S, Goei SL, van Veen K, Verhoeve N. Understanding teacher learning in lesson study through a cultural–historical activity theory lens. *Teach Teach Educ* 2022;119:103831. <https://doi.org/10.1016/j.tate.2022.103831>. Article 103831.
- [5] Alnemrat A, Aldamen H, Almashour M, Al-Deaibes M, AlShareef R. AI vs. teacher feedback on EFL argumentative writing: a quantitative study. *Front Educ* 2025;10. <https://doi.org/10.3389/educ.2025.1614673>. Article 1614673.
- [6] Ravi M, Besharat M. A holistic consideration of authentic assessments: student perception of assessment design, delivery, flexibility and creativity. *Eur J Eng Educ* 2026;51:25–42. <https://doi.org/10.1080/03043797.2025.2480116>.
- [7] Zarestky J, Bigler M, Brazile M, Lopes T, Bangerth W. Reflective writing supports metacognition and self-regulation in graduate computational science and engineering. *Comput Educ Open* 2022;3:100085. <https://doi.org/10.1016/j.caeo.2022.100085>. Article 100085.
- [8] Sasson I, Tifferet S, Lev-Ari L. Enhancing undergraduate metacognitive awareness and self-efficacy in research question formulation. *Eur J Educ* 2025;60(1). <https://doi.org/10.1111/ejed.12888>.
- [9] Abdi Tabari M. Investigating the interactions between L2 writing processes and products under different task planning time conditions. *J Second Lang Writ* 2022;55:100871. <https://doi.org/10.1016/j.jslw.2022.100871>.
- [10] Révész A, Michel M, Lu X, Kourtali N, Lee M, Borges L. The relationship of proficiency to speed fluency, pausing, and eye-gaze behaviours in L2 writing. *J Second Lang Writ* 2022;58:100927. <https://doi.org/10.1016/j.jslw.2022.100927>.
- [11] Gravett K. Authentic assessment as relational pedagogy. *Teach High Educ* 2025;30(3):608–22. <https://doi.org/10.1080/13562517.2024.2380997>.
- [12] Carless D, Boud D. The development of student feedback literacy: enabling uptake of feedback. *Assess Eval High Educ* 2018;43(8):1315–25. <https://doi.org/10.1080/02602938.2018.1463354>.
- [13] Mekheimer M. Generative AI-assisted feedback and EFL writing: a study on proficiency, revision frequency and writing quality. *Discov Educ* 2025;4. <https://doi.org/10.1007/s44217-025-00602-7>. Article 170.
- [14] Banihashem SK, Kerman NT, Noroozi O, Moon J, Drachsler H. Feedback sources in essay writing: peer-generated or AI-generated feedback? *Int J Educ Technol High Educ* 2024;21(1). <https://doi.org/10.1186/s41239-024-00455-4>. Article 23.
- [15] Bearman M, Tai J, Dawson P, Boud D, Ajjawi R. Developing evaluative judgement for a time of generative artificial intelligence. *Assess Eval High Educ* 2024;49(6):893–905. <https://doi.org/10.1080/02602938.2024.2335321>.
- [16] Schraw G, Dennison RS. Assessing metacognitive awareness. *Contemp Educ Psychol* 1994;19(4):460–75. <https://doi.org/10.1006/ceps.1994.1033>.
- [17] Zhan Y. Generative AI as enabler of student feedback engagement. *High Educ Res Dev* 2025;44(5):1289–304. <https://doi.org/10.1080/07294360.2025.2476513>.
- [18] Teng MF, Ma M. Assessing metacognition-based student feedback literacy for academic writing. *Assess Writ* 2024;59:100811. <https://doi.org/10.1016/j.asw.2024.100811>. Article 100811.

- [19] Wang H, Liu M, Zhang Y, Shu J. The usage of a transformer-based artificial intelligence-driven multidimensional feedback system in English writing instruction. *Sci Rep* 2025;15. <https://doi.org/10.1038/s41598-025-05026-9>.
- [20] Ajjawi R, Tai J, Dollinger M, Dawson P, Boud D, Bearman M. From authentic assessment to authenticity in assessment: broadening perspectives. *Assess Eval High Educ* 2024;49(4):499–510. <https://doi.org/10.1080/02602938.2023.2271193>.
- [21] Ba S, Yang L, Yan Z, Looi CK, Gašević D. Unraveling the mechanisms and effectiveness of AI-assisted feedback in education: a systematic literature review. *Comput Educ Open* 2025;9(9):100284. <https://doi.org/10.1016/j.caeo.2025.100284>. Article 100284.
- [22] Creswell JW, Creswell JD. *Research design: Qualitative, quantitative, and mixed methods approaches*. 6th ed. SAGE Publications; 2023.
- [23] Creswell JW, Plano Clark VL. *Designing and conducting mixed methods research*. 4th ed. SAGE; 2022.
- [24] OpenAI. GPT-3.5-turbo model card. <https://platform.openai.com/docs/models/gpt-3-5-turbo>; 2023.
- [25] García-Pérez MA. Use and misuse of corrections for multiple testing. *Methods Psychol* 2023;8:100120. <https://doi.org/10.1016/j.metip.2023.100120>. Article 100120.
- [26] Rubin M. Inconsistent multiple testing corrections: the fallacy of using family-based error rates to make inferences about individual hypotheses. *Methods Psychol* 2024;10:100140. <https://doi.org/10.1016/j.metip.2024.100140>. Article 100140.
- [27] Braun V, Clarke V. Using thematic analysis in psychology. *Qual Res Psychol* 2006;3(2):77–101. <https://doi.org/10.1191/1478088706qp0630a>.
- [28] Braun V, Clarke V. Toward good practice in thematic analysis: avoiding common problems and be(com)ing a knowing researcher. *Int J Transgender Health* 2023;24(1):1–6. <https://doi.org/10.1080/26895269.2022.2129597>.
- [29] Polit DF, Beck CT. *Nursing research: Generating and assessing evidence for nursing practice*. 11th ed. Wolters Kluwer; 2021.
- [30] White M, Ronfeldt M. Monitoring rater quality in observational systems: issues due to unreliable estimates of rater quality. *Educ Assess* 2024;29(2):124–46. <https://doi.org/10.1080/10627197.2024.2354311>.
- [31] Kang K, Jones MT, Armstrong K, Avery S, McHugo M, Heckers S, Vandekar S. Accurate confidence and Bayesian interval estimation for non-centrality parameters and effect size indices. *Psychometrika* 2023;88:253–73. <https://doi.org/10.1007/s11336-022-09899-x>.
- [32] Cohen J. *Statistical Power Analysis for the Behavioral Sciences*. 2nd ed. Hillsdale, NJ: Lawrence Erlbaum Associates; 1988.
- [33] Merkebu J, Kitsantas A, Durning SJ. What is metacognitive reflection? *Front Educ* 2023;8. <https://doi.org/10.3389/educ.2023.1166195>. Article 1166195.
- [34] Fetter MD, Tajima C. Joint displays of integrated data collection in mixed methods research. *Int J Qual Methods* 2022;21:1–13. <https://doi.org/10.1177/16094069221104564>.
- [35] Abdi Tabari M, Zhuang J, Farahanynia M. Task repetition and L2 written performance: a meta-analysis. *J Second Lang Writ* 2025;70:101255. <https://doi.org/10.1016/j.jslw.2025.101255>.
- [36] Liu J, Tang Y. The effect of task repetition on L2 learners' writing performance: a meta-analysis. *Lang Teach Res* 2025. <https://doi.org/10.1177/13621688251381268>.
- [37] Snijders TAB, Bosker RJ. *Multilevel analysis: An introduction to basic and advanced multilevel modeling*. 2nd ed. SAGE; 2012.